

[D.3.1] Universal AI: Solomonoff Induction

David Quarel

Iliad Intensive

June 2026

The big picture

- **Universal AI** asks: what would an *optimally intelligent* agent look like, given unlimited compute and the weakest possible assumptions?
- It splits into two problems:
 - 1 **Prediction (Solomonoff)**: passively observe a sequence $x_1, x_2, \dots$ and predict the next symbol. *How should you learn?*
 - 2 **Action (AIXI)**: interact with an unknown environment, take actions, collect reward. *How should you act?*
- Constructed as a single **Bayesian mixture** ξ over a class $\mathcal{M}$ of candidate environments, weighted by a prior w_ν .
 - Solomonoff: ξ (eventually) predicts as well as the truth μ .
 - AIXI: the Bayes-optimal policy π_ξ^* (eventually) acts as well as the optimal policy π_μ^* for true environment μ .
- Choose $\mathcal{M}$ = all computable environments, prior $w_\nu = 2^{-K(\nu)}$ (K is the “complexity” of the environment). Assumption “ $\mu \in \mathcal{M}$ ” becomes “*the universe is computable*” and the prior $w_\nu = 2^{-K(\nu)}$ encodes *Occam's razor* (simpler universes are more likely).

Formalization

- A true (unknown) environment μ generates a binary sequence; each bit $x_t \sim \mu(\cdot \mid x_{<t})$.
- A **predictor** P sees the past $x_{<t}$ and outputs a distribution $P(\cdot \mid x_{<t})$ over the next bit (next-token prediction)
- **Quality measure:** Cumulative μ -expected squared prediction error
 $S_\infty^\mu := \sum_{t=1}^{\infty} s_t^\mu$

Per-step squared prediction error s_t^μ

$$s_t^\mu = \sum_{x_{<t}} \mu(x_{<t}) \left[\sum_{x_t} (P(x_t \mid x_{<t}) - \mu(x_t \mid x_{<t}))^2 \right]$$

- **Goal:** build a P that makes cumulative error S_∞^μ *small*.
- **Idea:** don't bet on one model. Hedge your bets and mix over a whole class $\mathcal{M} = \{\nu_1, \nu_2, \dots\}$ of potential environments.

The Bayesian mixture ξ

Definition

Class $\mathcal{M} = \{\nu_1, \nu_2, \dots\}$ with prior weights $w_\nu > 0$, $\sum_\nu w_\nu = 1$.

$$\xi(x_{1:t}) := \sum_{\nu \in \mathcal{M}} w_\nu \nu(x_{1:t}).$$

- **Key fact (mixture is a predictor):** its one-step prediction is a *posterior-weighted* average of the members' predictions (you prove this today!)

$$\xi(x_t \mid x_{<t}) = \sum_{\nu \in \mathcal{M}} w(\nu \mid x_{<t}) \nu(x_t \mid x_{<t}), \quad w(\nu \mid x_{<t}) = w_\nu \frac{\nu(x_{<t})}{\xi(x_{<t})}.$$

- **Posterior update is multiplicative:** $w(\nu \mid x_{1:t}) = w(\nu \mid x_{<t}) \cdot \frac{\nu(x_t \mid x_{<t})}{\xi(x_t \mid x_{<t})}$:
we reweight by how well ν predicted vs. ξ (Bayes' rule!)

Main result: the cumulative prediction bound

Cumulative bound

$$S_{\infty}^{\mu} := \sum_{t=1}^{\infty} \sum_{x_{<t}} \mu(x_{<t}) \sum_{x_t \in \mathbb{B}} (P(x_t \mid x_{<t}) - \mu(x_t \mid x_{<t}))^2 \leq -\ln w_{\mu}$$

- Total squared error over *all time* is finite, bounded by the prior penalty on μ .
- Higher prior on the truth implies smaller total error.
- There are infinitely many environments. Cannot choose a uniform prior.
What to choose?

Specializing to the Solomonoff prior

- Take the **Solomonoff prior** $w_\nu = 2^{-K(\nu)}$, where $K(\nu)$ is the *Kolmogorov complexity*: length of the shortest program computing ν .
- Substitute $w_\mu = 2^{-K(\mu)}$ into the cumulative bound:

Explicit complexity bound

$$S_\infty^\mu \leq K(\mu) \ln 2$$

- **Interpretation:** the total prediction error is (roughly) bounded by the **complexity** of the true environment. Simpler worlds are learned faster.
- This is the formal version of **Occam's razor**.

Other properties of the Solomonoff prior

- **Bounded mistakes:** on a deterministic sequence, predicting the most likely bit $\hat{x}_t = \arg \max_{x \in \mathbb{B}} \xi(x \mid x_{<t})$ makes at most $-2 \ln w_\mu$ wrong guesses *ever*.
- **Pareto optimality:** no other predictor beats ξ on *every* $\nu \in \mathcal{M}$ simultaneously (for KL loss *and* squared loss).
- **Misspecification** ($\mu \notin \mathcal{M}$): the bound degrades gracefully,

$$S_n^\mu \leq -\ln w_{\hat{\mu}} + D_n(\mu \parallel \hat{\mu})$$

where $\hat{\mu}$ is the closest model in $\mathcal{M}$ to μ .

Why care? Applications, approximations, critiques

- **A yardstick for prediction.** Solomonoff's bound [Sol78] is tight up to additive $\sqrt{\cdot}$ terms [Hut01] and Pareto-optimal for general losses [Hut03]; it dissolves classic puzzles of Bayesian confirmation (zero-prior, black ravens) [Hut07] and grounds a Bayesian theory of induction [RH11].
- **Computable cousins that work.** Shrink $\mathcal{M}$ and the same mixture becomes an algorithm: context tree weighting [WST95] is the exact Bayes mixture over all variable-order Markov models, in linear time; MDL [Grü07] replaces the mixture by its best single code; the speed prior [Sch02] trades universality for computability.
- **Prediction is compression.** The Hutter Prize scores intelligence by compressed size of Wikipedia; LLMs turn out to be strong general-purpose compressors [DRD⁺24].
- **Deep learning through this lens.** Meta-training nets on UTM-generated data amortises Solomonoff induction [GMGH⁺24]; transformers conjectured as its best extant approximation [YW25]; K -complexity explains the simplicity bias behind generalisation [GFRW24, MRVPL25].
- **The critiques.** No elegant computable universal predictor exists [Leg06]; different UTMs give non-equivalent priors [WSH13]; the “Occam's razor” reading is contested: the real assumption is computability of the world [Ste16, Ste26].

Takeaways

- **Solomonoff** solves induction: mix over all computable hypotheses, weighted by simplicity. Total prediction error $\leq K(\mu) \ln 2$.
- **AIXI** lifts this to action: the Bayes-optimal policy over the same universal mixture. It learns to predict *and* to act as well as if it knew the world.
- **The catch:** both are *incomputable* (K is incomputable; $\mathcal{M}$ is intractable to enumerate). They are gold standards, not computable algorithms, and the prior's $2^{-K(\nu)}$ hides a dependency on the choice of universal Turing machine.

References I

- [DRD⁺24] Grégoire Delétang, Anian Ruoss, Paul-Ambroise Duquenne, Elliot Catt, Tim Genewein, Christopher Mattern, Jordi Grau-Moya, Li Kevin Wenliang, Matthew Aitchison, Laurent Orseau, Marcus Hutter, and Joel Veness. Language modeling is compression. In *International Conference on Learning Representations (ICLR)*, 2024. URL: <https://arxiv.org/abs/2309.10668>.
- [GFRW24] Micah Goldblum, Marc Finzi, Keefer Rowan, and Andrew Gordon Wilson. Position: The no free lunch theorem, Kolmogorov complexity, and the role of inductive biases in machine learning. In *International Conference on Machine Learning (ICML)*, 2024. URL: <https://arxiv.org/abs/2304.05366>.
- [GMGH⁺24] Jordi Grau-Moya, Tim Genewein, Marcus Hutter, Laurent Orseau, Grégoire Delétang, Elliot Catt, Anian Ruoss, Li Kevin Wenliang, Christopher Mattern, Matthew Aitchison, and Joel Veness. Learning universal predictors. In *International Conference on Machine Learning (ICML)*, 2024. URL: <https://arxiv.org/abs/2401.14953>.
- [Grü07] Peter D. Grünwald. *The Minimum Description Length Principle*. MIT Press, 2007. URL: <https://direct.mit.edu/books/monograph/3813/The-Minimum-Description-Length-Principle>.
- [Hut01] Marcus Hutter. New error bounds for Solomonoff prediction. *Journal of Computer and System Sciences*, 62(4):653–667, 2001. URL: <http://www.hutter1.net/ai/perrbnd.htm>.
- [Hut03] Marcus Hutter. Optimality of universal Bayesian sequence prediction for general loss and alphabet. *Journal of Machine Learning Research*, 4:971–1000, 2003. URL: <https://www.jmlr.org/papers/v4/hutter03a.html>.
- [Hut07] Marcus Hutter. On universal prediction and Bayesian confirmation. *Theoretical Computer Science*, 384(1):33–48, 2007. URL: <https://arxiv.org/abs/0709.1516>.
- [Leg06] Shane Legg. Is there an elegant universal theory of prediction? In *Algorithmic Learning Theory (ALT 2006)*, volume 4264 of *Lecture Notes in Computer Science*, pages 274–287. Springer, 2006. URL: <https://arxiv.org/abs/cs/0606070>.
- [MRVPL25] Chris Mingard, Henry Rees, Guillermo Valle-Pérez, and Ard A. Louis. Deep neural networks have an inbuilt Occam's razor. *Nature Communications*, 16:220, 2025. doi:10.1038/s41467-024-54813-x.
- [RH11] Samuel Rathmanner and Marcus Hutter. A philosophical treatise of universal induction. *Entropy*, 13(6):1076–1136, 2011. URL: <https://arxiv.org/abs/1105.5721>.
- [Sch02] Jürgen Schmidhuber. The speed prior: A new simplicity measure yielding near-optimal computable predictions. In *Computational Learning Theory (COLT 2002)*, volume 2375 of *Lecture Notes in Computer Science*, pages 216–228. Springer, 2002. URL: https://link.springer.com/chapter/10.1007/3-540-45435-7_15.
- [Sol78] R. J. Solomonoff. Complexity-based induction systems: Comparisons and convergence theorems. *IEEE Transactions on Information Theory*, 24(4):422–432, 1978. URL: <https://raysolomonoff.com/publications/solo1.pdf>.

References II

- [Ste16] Tom F. Sterkenburg. Solomonoff prediction and Occam's razor. *Philosophy of Science*, 83(4):459–479, 2016. doi:[10.1086/687257](https://doi.org/10.1086/687257).
- [Ste26] Tom F. Sterkenburg. Solomonoff induction. arXiv:2603.20274, 2026. URL: <https://arxiv.org/abs/2603.20274>.
- [WSH13] Ian Wood, Peter Sunehag, and Marcus Hutter. (Non-)Equivalence of universal priors. In *Algorithmic Probability and Friends: Bayesian Prediction and Artificial Intelligence (Solomonoff 85th Memorial Conference)*, volume 7070 of *Lecture Notes in Computer Science*, pages 417–425. Springer, 2013. URL: <https://arxiv.org/abs/1111.3854>.
- [WST95] F. M. J. Willems, Y. M. Shtarkov, and T. J. Tjalkens. The context-tree weighting method: Basic properties. *IEEE Transactions on Information Theory*, 41(3):653–664, 1995. doi:[10.1109/18.382012](https://doi.org/10.1109/18.382012).
- [YW25] Nathan Young and Michael Witbrock. Transformers as approximations of Solomonoff induction. In *Neural Information Processing (ICONIP 2024)*. Springer, 2025. URL: <https://arxiv.org/abs/2408.12065>.